

Solution Requirements

Date	03 July 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Session state user validation; transition to JWT tokens in Phase 2.	Medium
2	Authorization levels	Access levels for Student (basic modules) and Admin/Educator (backend adjustments).	Medium
3	External interfaces	Integration with Google Gemini 2.5 Flash API and Hugging Face Hub.	High
4	Transactions processing	Serializing interactive quizzes to strict JSON blocks over POST endpoints.	High
5	Reporting	Rendering quiz performance metrics and exporting markdown summarizations.	High
6	Business rules, etc.	Running heavy tasks via cloud APIs; routing explanations to local AI models.	High

7	Compliance to laws or regulations	Strict transient, in-memory data processing to secure student academic privacy.	High
8	<i>Other</i>	Multi-page Streamlit grid UI with native header elements hidden.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Fast output responses.	Cloud tasks: < 3s. Local inferences: < 15s.
2	Scalability	Handle simultaneous student sessions.	Support up to 50 concurrent requests on local CPU.
3	Security & Data Privacy	Protected API endpoints.	Load keys via .env files; enforce character limits (< 5,000).
4	Reliability & Availability	Keep services active.	99.5% API uptime with graceful UI fallback error handling.
5	Usability & Accessibility	Legible and user-friendly interface	WCAG-compliant high contrast; all modules reachable in < 2 clicks.
6	<i>Other</i>	Run across setups.	Cross-platform Docker containerization using Python 3.10+.